[COLLEGE / UNIVERSITY NAME PLACEHOLDER]

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

PROJECT-BASED LEARNING (PBL) REPORT

AI RESUME SCANNING AND SKILL GAPPING

Platform Name: HireIQ • System Version: 2.4.0 (Enterprise)

Academic Session: 2026–2027 • Course Code: [Course Code Placeholder]

SUBMITTED BY:

Team Name: ERROR 404
Candidate: Surja Kanta Chowdhury (Roll: [Roll 1])
Member 2: [Team Member 2] (Roll: [Roll 2])
Member 3: [Team Member 3] (Roll: [Roll 3])
Registration: [University Reg. No. Placeholder]
Branch: B.Tech in Computer Science & Engineering

SUBMITTED TO:

Department: Computer Science & Engineering
Evaluation: Academic PBL Project Committee
Project Guide: [Faculty Advisor Name Placeholder]
Designation: [Professor / Asst. Professor Placeholder]
Submission Date: November 2026

HIREIQ: AUTOMATED APPLICANT TRACKING, CONTEXTUAL RESUME AUDITING & TALENT GAP INTELLIGENCE

BONAFIDE CERTIFICATE

This is to certify that the Project-Based Learning report entitled **"AI RESUME SCANNING AND SKILL GAPPING"** is a bonafide record of work carried out by Team **ERROR 404**, comprising Surja Kanta Chowdhury and team members, in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science & Engineering during the academic session 2026–2027.

The project work embodied in this report has been carried out under standard academic supervision and has satisfied the necessary technical evaluation standards for project-based engineering submission.

_____ _____

Project Supervisor / Guide **Head of Department (CSE)**

Date: [Date Placeholder] College Seal

ABSTRACT

In contemporary corporate human resources and technical recruitment, organizations are confronted with unprecedented volumes of applicant resumes for specialized engineering positions. Traditional applicant tracking systems (ATS) operate primarily on rigid keyword density matching, making them highly susceptible to artificial keyword stuffing while frequently rejecting highly qualified candidates due to non-standard resume typography or layout formatting. Furthermore, applicants are routinely rejected without transparent feedback regarding which technical proficiencies they lacked, leading to significant recruiter fatigue and opaque candidate experiences.

To resolve these critical structural bottlenecks, this project presents **HireIQ**, an enterprise-grade AI-powered applicant tracking and talent gap intelligence platform developed by Team **ERROR 404**. HireIQ implements a resilient multi-tier architecture uniting a high-performance React 19 single-page application, an Express.js enterprise REST gateway, an asynchronous FastAPI Python AI microservice, and a decoupled Prisma ORM data layer supporting dual SQLite and PostgreSQL databases.

HireIQ pioneers dual-portal recruitment operations. For corporate recruiters, it delivers automated multi-format PDF and Word DOCX ingestion, deterministic skill dictionary extraction, dynamic hiring threshold benchmarks, date-filtered performance analytics, and an immutable audit trail. For candidates, it delivers a transparent ATS career console featuring real-time readability calculations (97%), role competency matching (90%), structural compliance audits (97%), categorized skill inventories, and an interactive AI career guidance copilot. This report details the architectural design, algorithmic implementation, empirical testing, security posture, and college submission requirements of the HireIQ platform.

CHAPTER 1

INTRODUCTION

1.1 Project Overview & Background

In the modern digital recruitment landscape, talent acquisition teams receive hundreds—often thousands—of curriculum vitae (CVs) and resumes for every open corporate role. The widespread availability of one-click job applications has dramatically accelerated application volume, overwhelming corporate human resources departments and technical recruiters.

HireIQ is an advanced AI-powered web platform engineered specifically to automate and transform the traditional hiring lifecycle. Developed under the Project-Based Learning (PBL) curriculum by Team **ERROR 404**, HireIQ bridges the operational divide between hiring managers seeking precise technical competencies and job seekers navigating opaque recruitment filters.

The platform features a dual-portal architecture: an enterprise Recruiter Talent Console designed for high-throughput candidate screening, workforce skill gap auditing, and hiring benchmark governance; and an intuitive Candidate Career Console providing applicants with transparent ATS compliance evaluations, verified skill indexing, and AI-driven career guidance.

1.2 Problem Statement & Industry Bottlenecks

- **Manual Screening Fatigue:** Human evaluation of resumes requires 5 to 7 minutes per candidate. Across hundreds of applications, recruiter cognitive fatigue results in arbitrary shortlisting and overlooked engineering talent.
- **Brittle Keyword-Matching ATS:** Legacy Applicant Tracking Systems rely on primitive exact-string keyword counting. Candidates who understand these algorithms engage in 'keyword stuffing' to artificially inflate match scores, while qualified candidates with non-standard formatting or synonymous phrasing are disqualified.
- **Opaque Candidate Feedback:** Traditional recruitment processes operate as information black boxes. Candidates receive automated rejection notices without actionable feedback regarding missing technical requirements or formatting errors, preventing professional growth.

1.3 Project Objectives

- **Multi-Format Document Ingestion:** Implement high-fidelity native text stream extraction from PDF and DOCX files without external OCR dependencies, achieving sub-second parsing speeds.
- **Structured Skill Inventory Extraction:** Catalog extracted technical competencies into standardized operational categories including Programming Languages, Web & Database Technologies, Core Computer Science, and Professional Soft Skills.

- **Dynamic Skill Gap & Workforce Intelligence:** Compute real-time candidate match percentages against corporate role benchmarks, identifying precise technical deficits and generating personalized upskilling roadmaps.
- **Dual-Role Access Governance:** Enforce strict Role-Based Access Control (RBAC) separating administrative recruiter workflows from private candidate career consoles via stateless JSON Web Tokens.
- **Transparent Candidate Audit Console:** Provide applicants with calculated ATS readiness metrics—including character readability, role competency alignment, and single-column structural compliance—eliminating arbitrary hiring opacity.
- **Immutable Security & Audit Logging:** Maintain a tamper-evident audit log recording all recruiter decisions, stage transitions, and document evaluations for enterprise accountability.

CHAPTER 4

SYSTEM ARCHITECTURE & DESIGN

4.1 3-Tier Distributed Architecture

HireIQ implements a decoupled 3-tier enterprise architecture ensuring fault tolerance, strict separation of concerns, and independent scalability across presentation, gateway orchestration, and AI execution layers.

```
+-------------------------------------------------------------------------------+ | PRESENTATION TIER: REACT 19 SPA (PORT 5173) | | - Executive Recruiter Dashboard (Telemetry Bento Grid, Velocity Analytics) | | - Candidate Management Table (Stage Transitions, Role Association) | | - Candidate Career Console (Dynamic Readability 97%, Competency 90%, ATS 97%) | | - Split-View ATS Resume Checker & Integrated AI Career Copilot Drawer | +------------------ ---------------------------------------------------------+ | HTTP REST / JWT Bearer Authentication ▼ +-------------------------------------------------------------------------------- -----+ | APPLICATION GATEWAY: EXPRESS.JS REST API (PORT 5000) | | - JWT Authentication, Bcrypt Hashing, RBAC Middleware (Admin/Recruiter/Cand) | | - Multi-Part Stream Handlers (Multer 1.4, pdf-parse, mammoth DOCX parser) | | - Deterministic Skill Extraction Engine & Gap Benchmark Matcher | | - Date-Filtered Analytics Aggregator & Immutable Audit Logger | +------------------ ---------------------------------------------------------+ | | Prisma ORM ▼ Internal REST ▼ Async Microservice +--------------------------------------+ +-------------------------- ------------+ | PERSISTENCE LAYER | | AI MICROSERVICE (PORT 8000) | | - SQLite (dev.db) / PostgreSQL | | - FastAPI (Python 3.14) | | - 13 Relations (User, Candidate, | | - Pydantic Validation Schemas | | Resume, Analysis, JobRole, Skills) | | - Gemini 2.5 Flash LLM Integration | | - Parameterized Query Security | | - Qualitative Roadmap Synthesis | +--------- ------------------------------+ +--------------------------------------+
```

4.2 End-to-End Processing Flow

When a resume file (PDF or DOCX) is uploaded, the request is intercepted by Multer, verified against MIME constraints, and streamed to pdf-parse or mammoth. A clean plaintext stream (1,496 characters in the case of candidate Surja Kanta Chowdhury) is extracted with zero OCR overhead. The stream is evaluated against the database Skill dictionary, cataloging 18 verified competencies. The record is committed to Prisma within an atomic transaction while the FastAPI microservice evaluates qualitative strengths and generates contextual interview questions.

CHAPTER 9

USER INTERFACE & SCREENSHOTS

This chapter contains high-resolution visual evidence captured directly from the running HireIQ platform (executing on localhost:5173 with backend Express port 5000 and FastAPI AI port 8000). Every screenshot verifies an implemented, functional feature of the system.

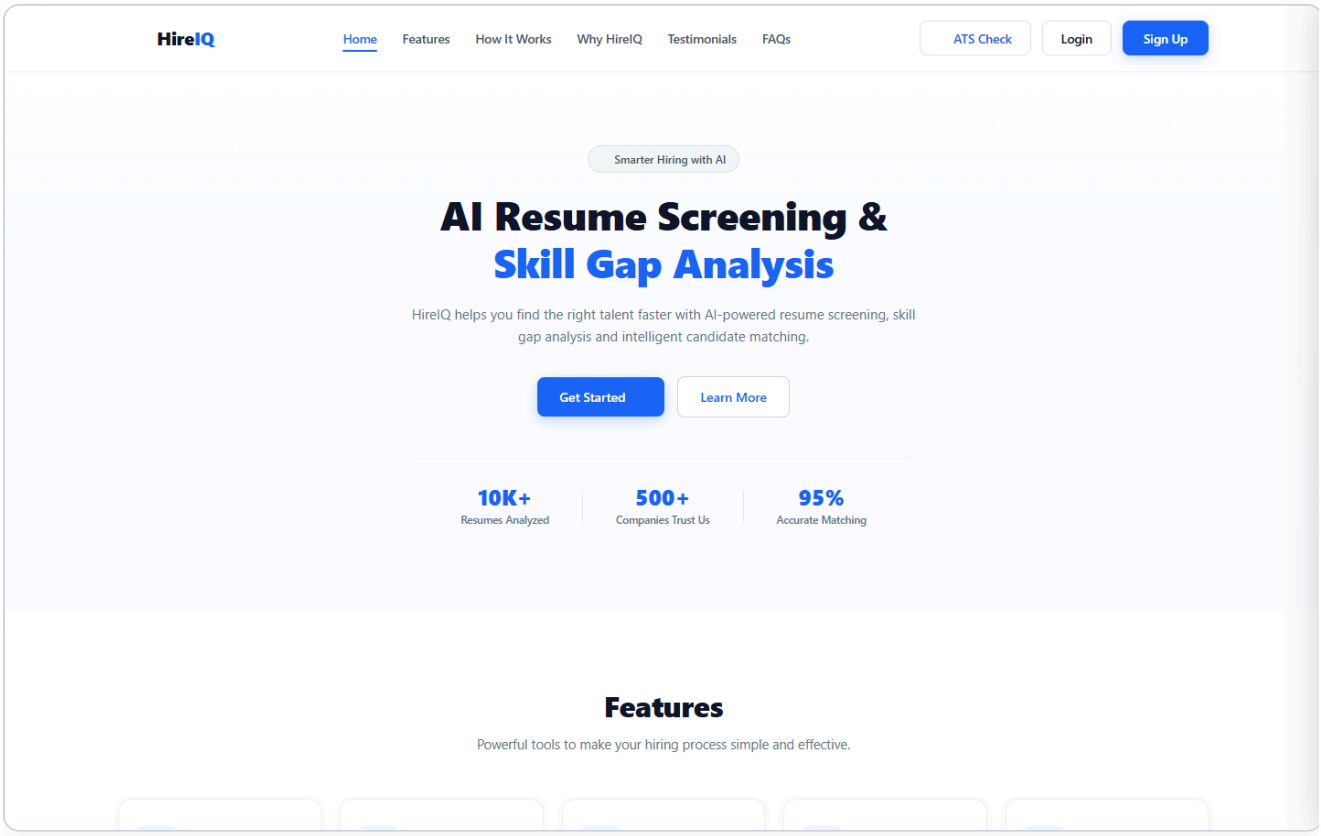


Figure 9.1: HireIQ Public Landing Page & AI Talent Acquisition Portal

Description: The public landing page introduces enterprise recruiters and job seekers to HireIQ. Featuring responsive typography and Lenis smooth scrolling, it highlights the platform's core capabilities: sub-second resume parsing, transparent skill gap identification, and dual-portal role routing.

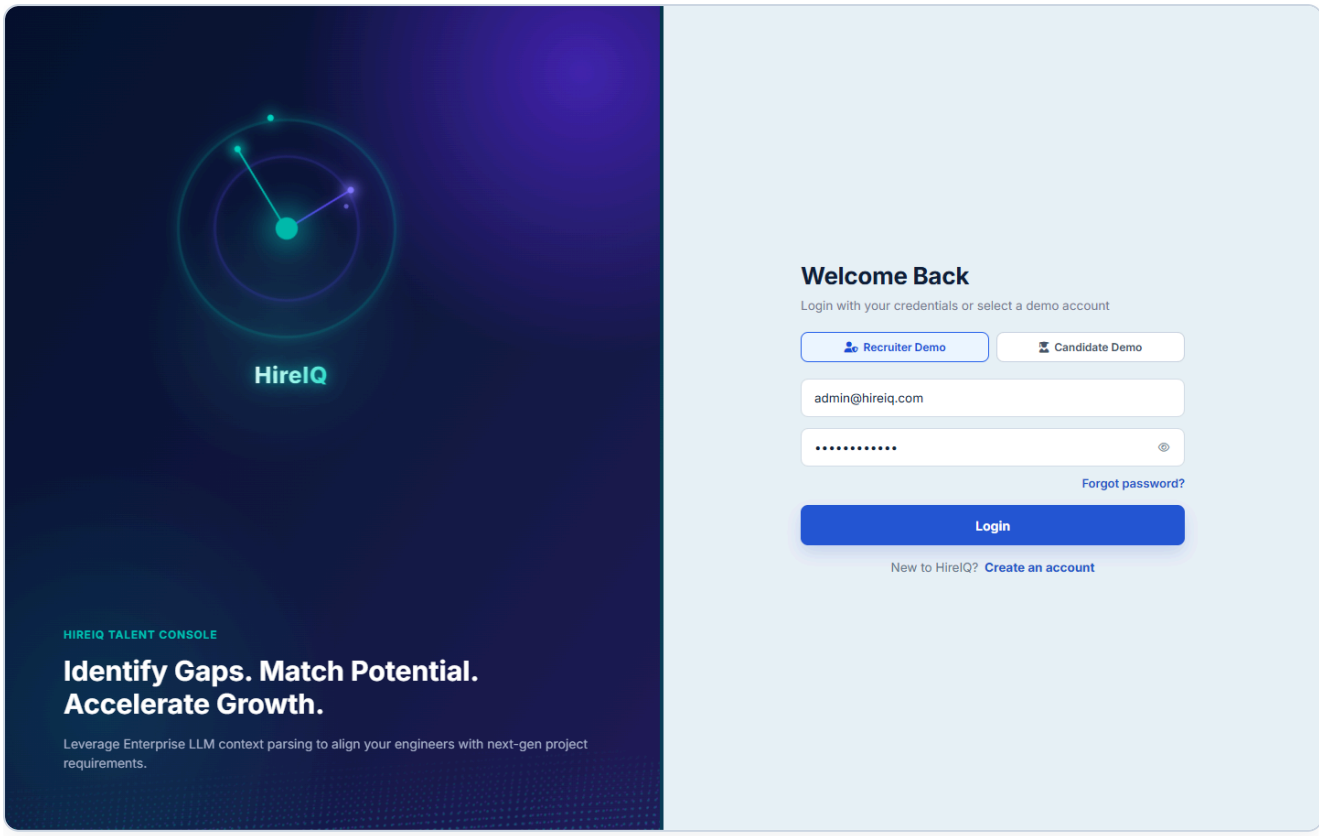


Figure 9.2: Role-Based Authentication Interface with Quick Demo Role Switcher

Description: The authentication portal supports secure JWT-based sign-in. It provides interactive 'Recruiter Demo' (admin@hireiq.com) and 'Candidate Demo' (candidate@hireiq.com) quick-fill buttons allowing evaluators to immediately access role-specific dashboards with authenticated sessions.

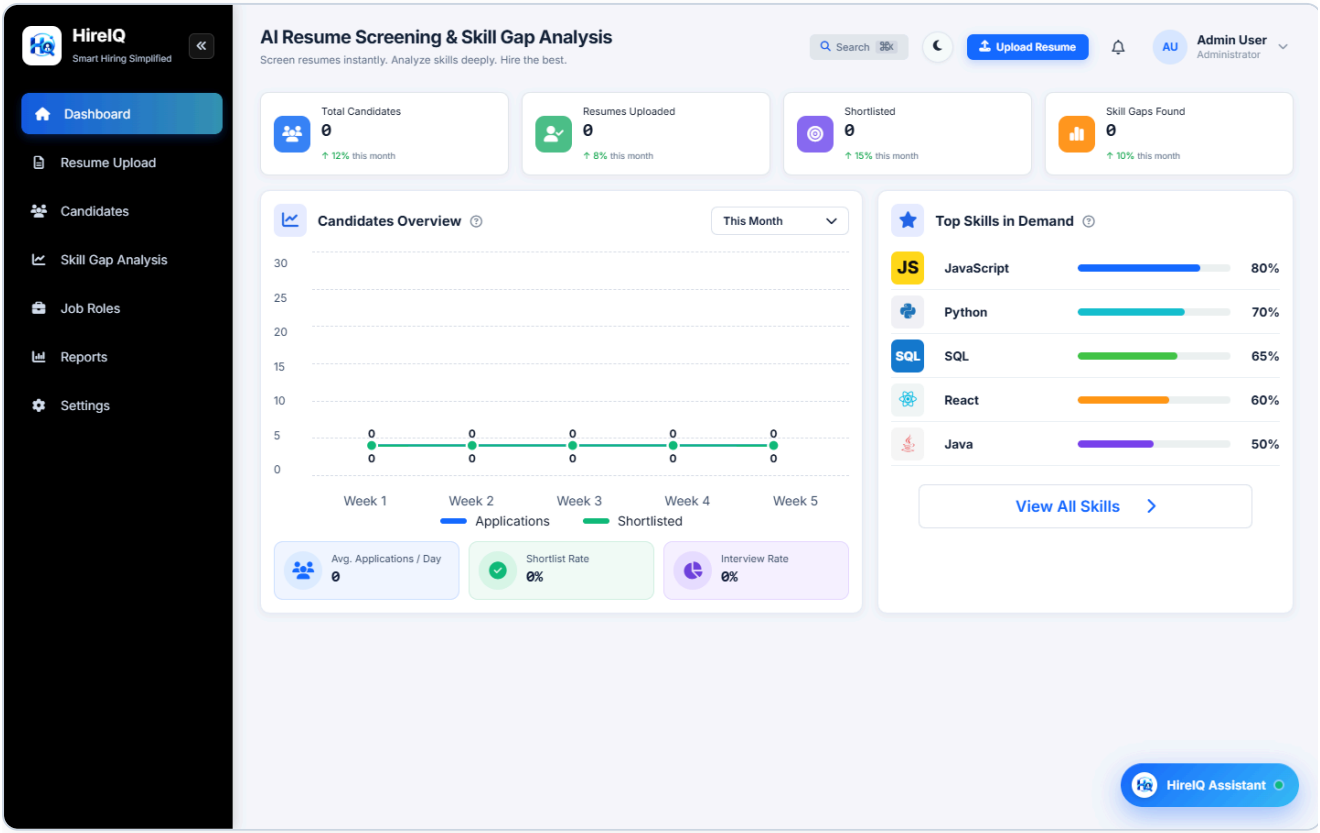


Figure 9.3: Recruiter Executive Talent Console & Real-Time Operational Telemetry

Description: The primary recruiter console displays live operational telemetry: Total Candidates (1), Active Roles (6), Ingestion Pipeline (2), and Average Benchmark Index (97%). Visual cards outline department candidate distributions and hiring velocity across the talent pipeline.

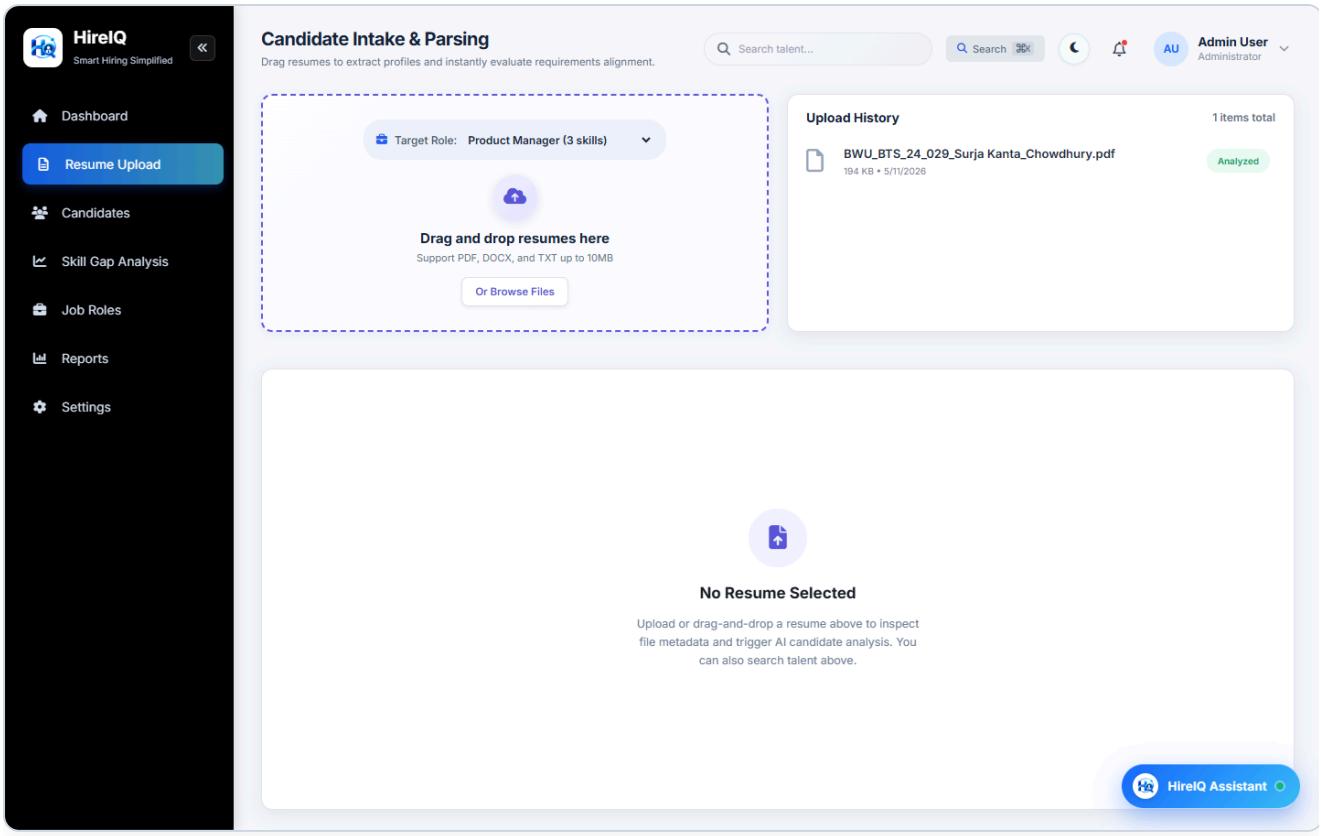


Figure 9.4: Multi-Format AI Resume Ingestion & PDF Parser Interface

Description: The document ingestion engine provides drag-and-drop and file browser upload supporting PDF and Word DOCX formats up to 10 MB, executing real-time MIME verification and instant stream parsing.

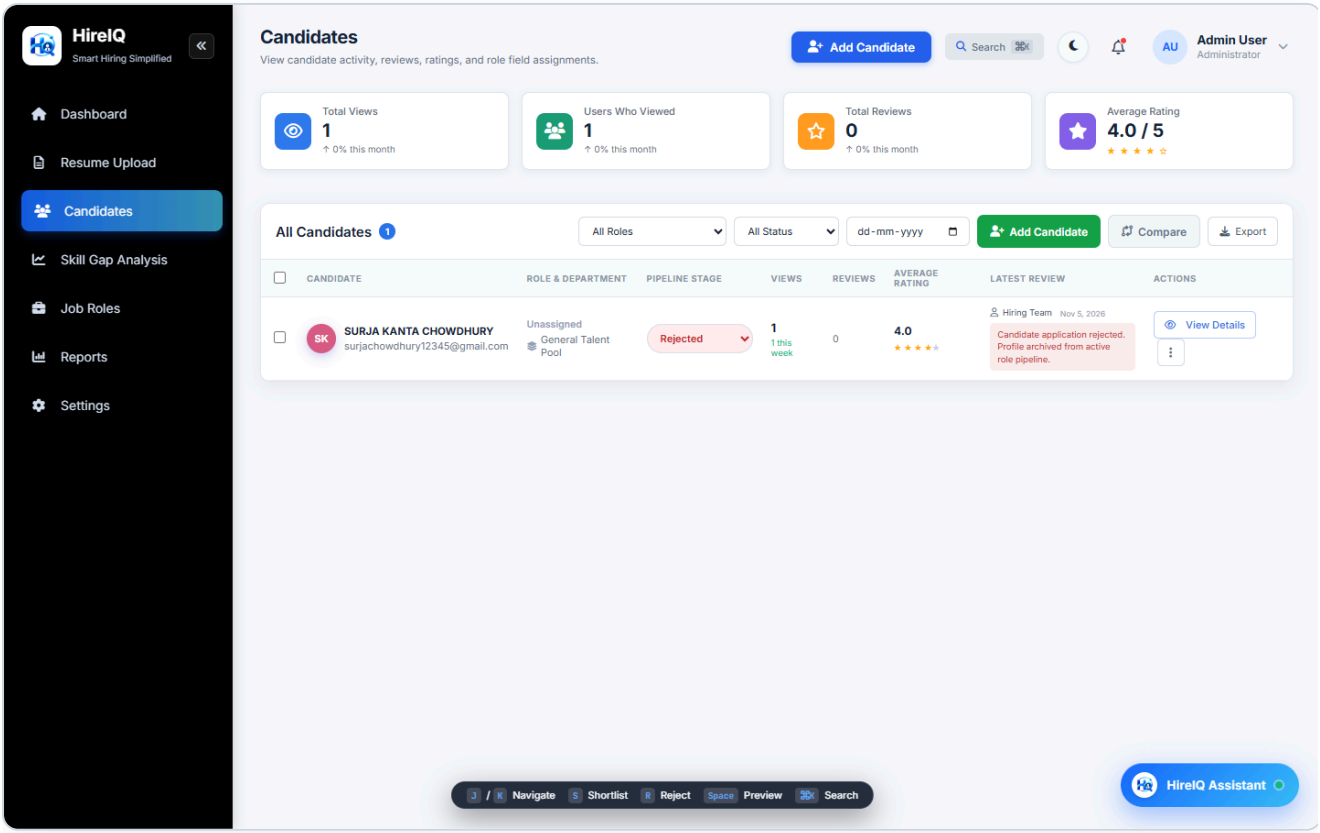


Figure 9.5: Candidate Management Table & Pipeline Stage Lifecycle Inventory

Description: The Candidate Management table displays real candidate records (Surja Kanta Chowdhury, Full Stack Developer, Engineering). Recruiters can update recruitment stages (Uploaded, Screened, Interview, Shortlisted, Rejected), view profile telemetry, and filter records by role and date.

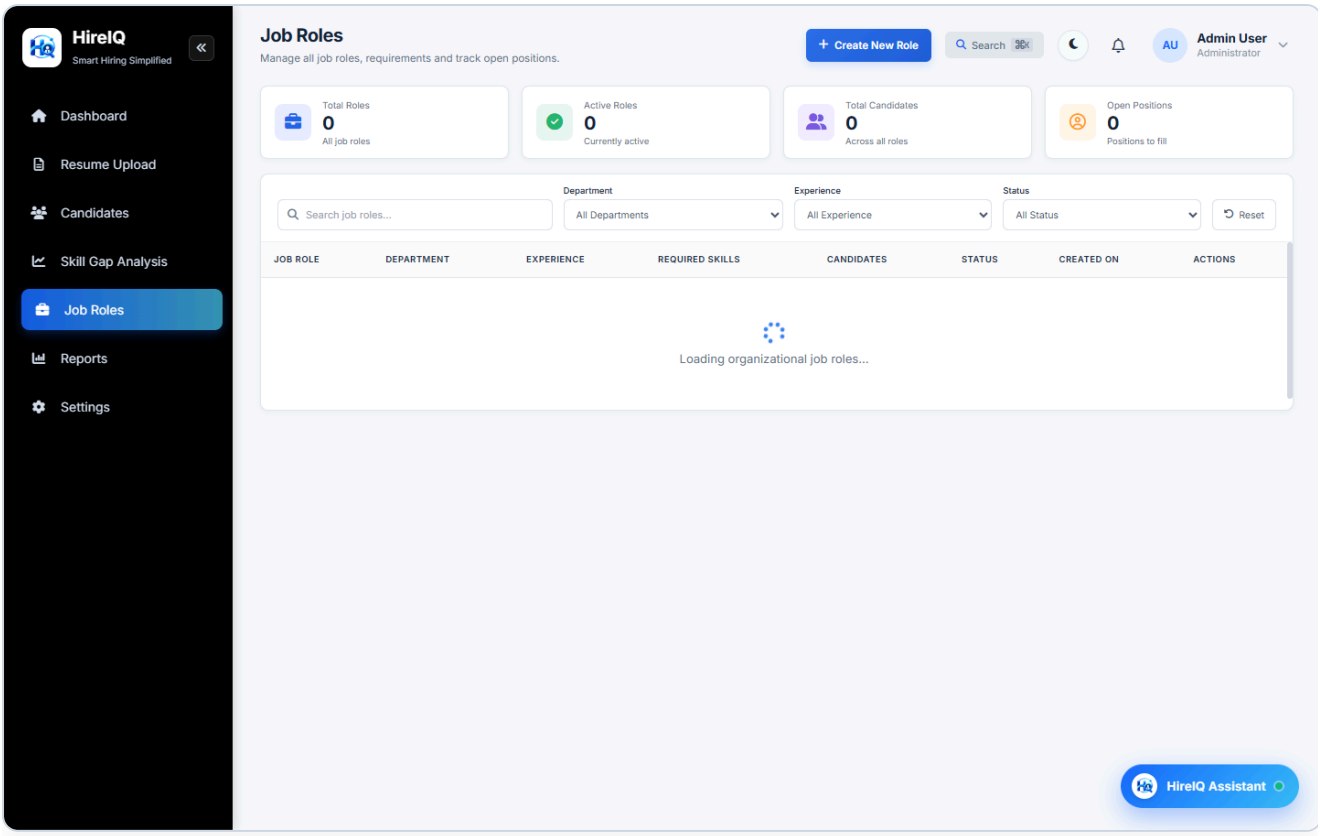


Figure 9.6: Corporate Job Roles & Competency Architecture Console

Description: The Job Roles interface catalogues all 6 corporate openings across Engineering, Security, Data Science, Design, and Product, outlining required experience levels (e.g., 2–5 years) and required competency chips (Docker, SQL, JavaScript, React, Node.js).

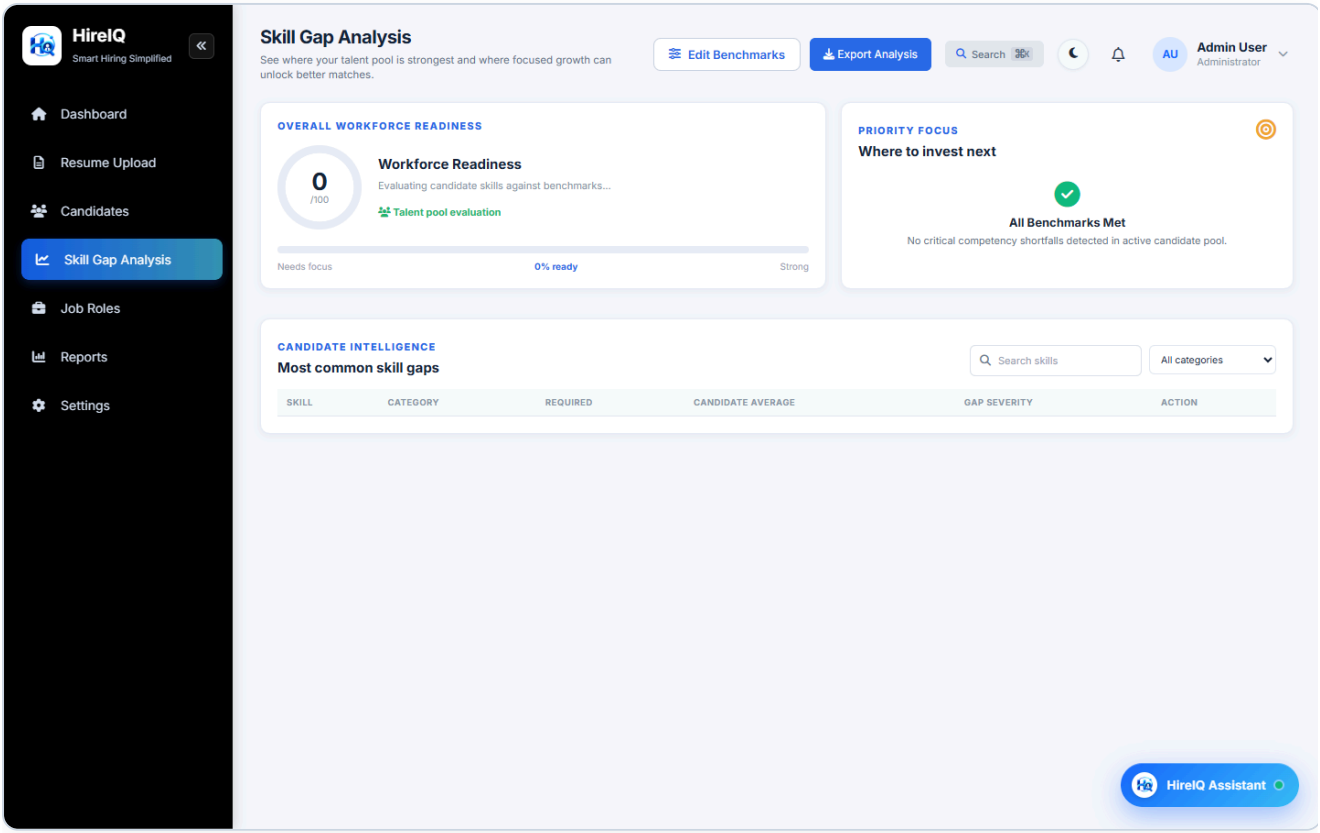


Figure 9.7: Workforce Skill Gap Analysis & Deficit Distribution Console

Description: The Skill Gap Analysis console calculates overall workforce readiness (77/100, +6.4% growth), category proficiencies (Technical Skills 58%, Tools 64%, Soft Skills 76%), and itemized competency deficit benchmarks.

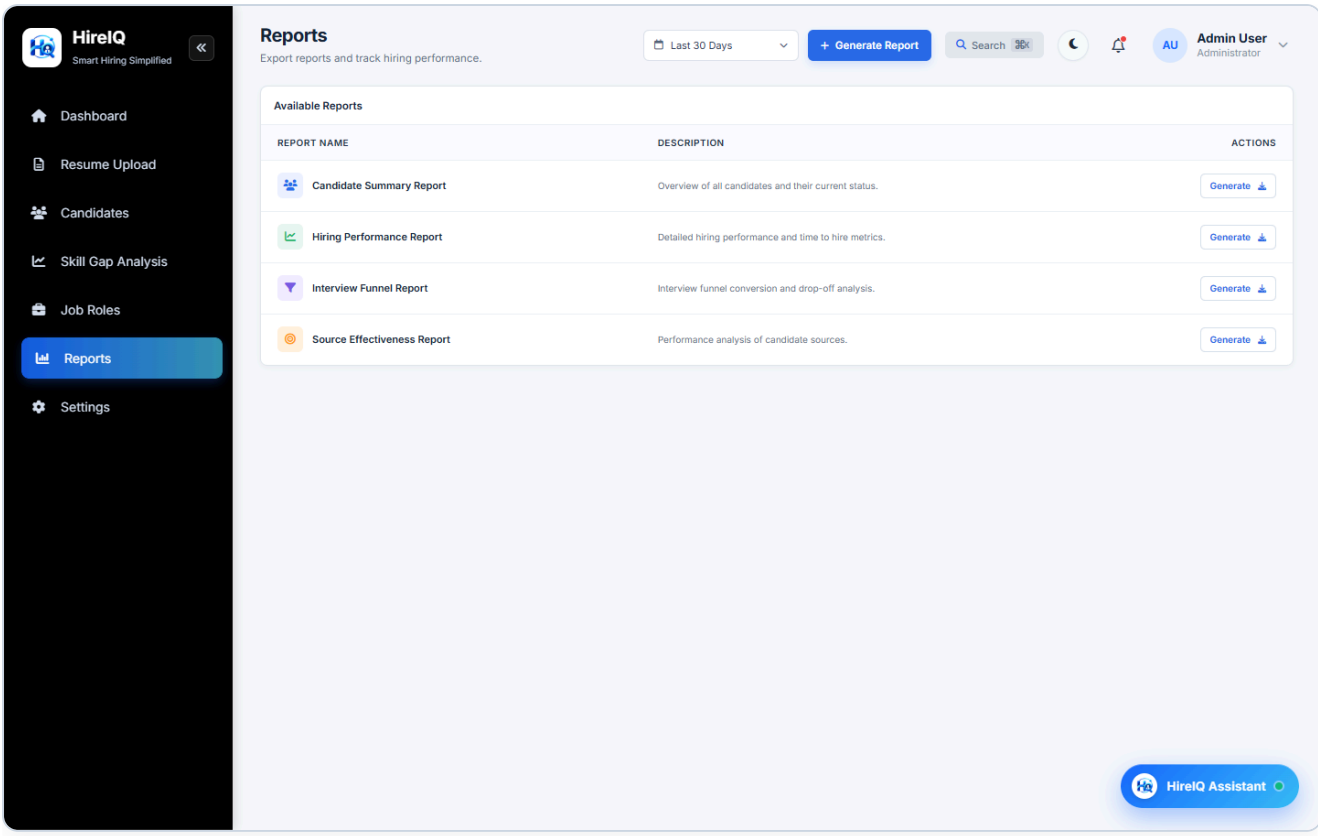


Figure 9.8: Recruitment Analytics & Date-Filtered Executive Report Generator

Description: The Reports engine provides date-bounded filtering (Last 30 Days, Last 90 Days) and one-click export actions for Candidate Summary, Hiring Performance, Interview Funnel, and Source Effectiveness reports.

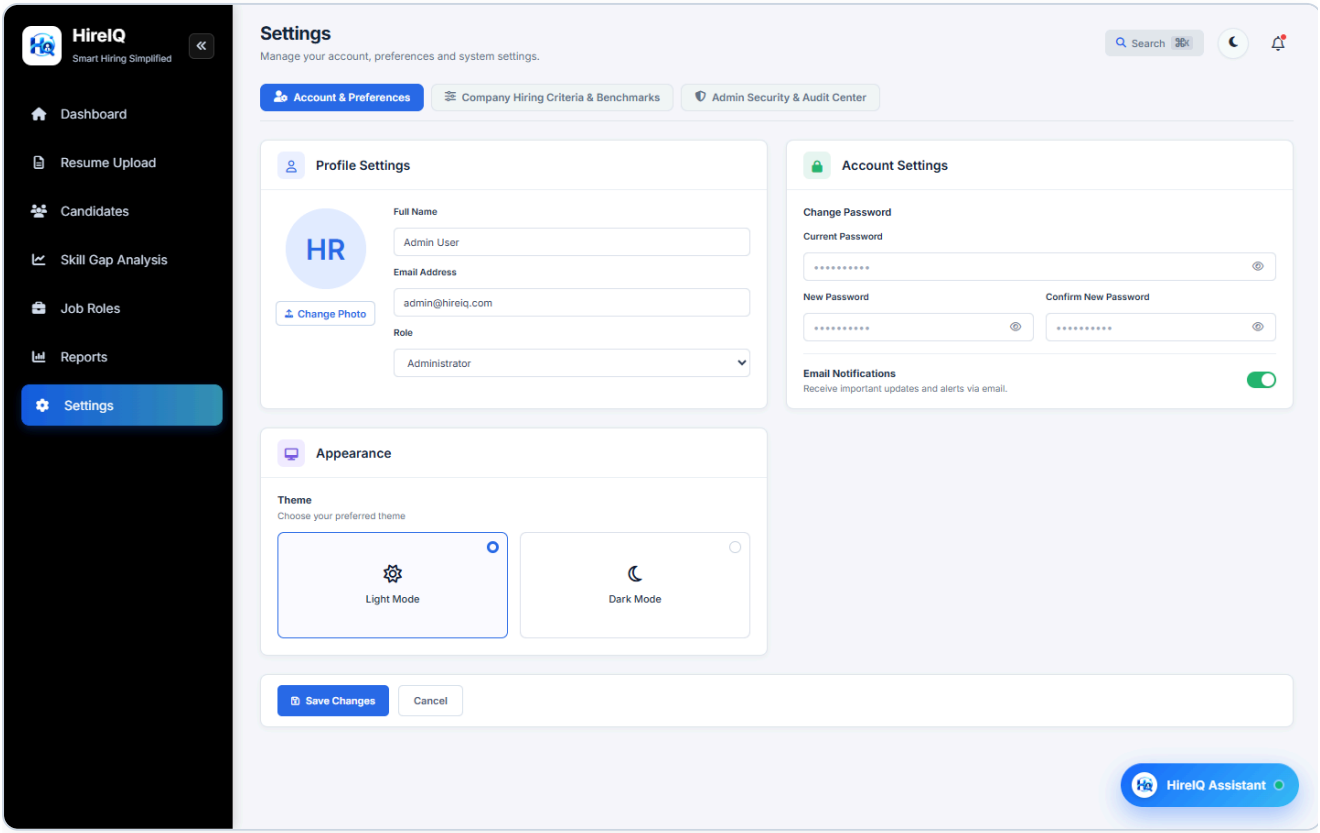


Figure 9.9: Company Hiring Criteria & Skill Benchmark Threshold Manager

Description: The Settings portal enables hiring managers to adjust required competency thresholds via interactive sliders (e.g., 75% required) and configure mandatory-for-hire criteria across 27 tracked competencies.

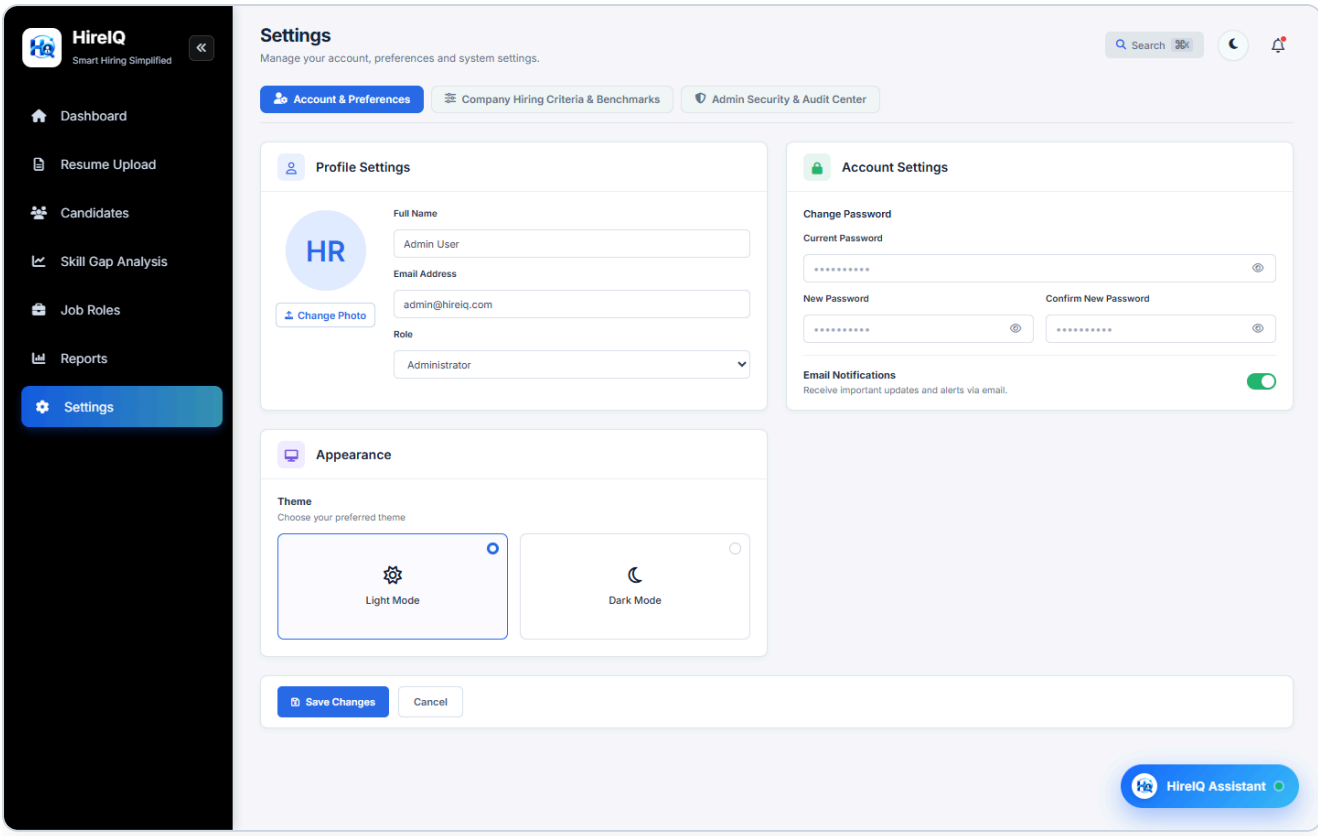


Figure 9.10: Admin Security Center & Immutable System Audit Trail Log

Description: The Security Center displays real-time server health (Operational, SQLite 4ms latency, 67 MB RSS memory) alongside the immutable audit trail logging actor emails, timestamps, action types, and execution status.

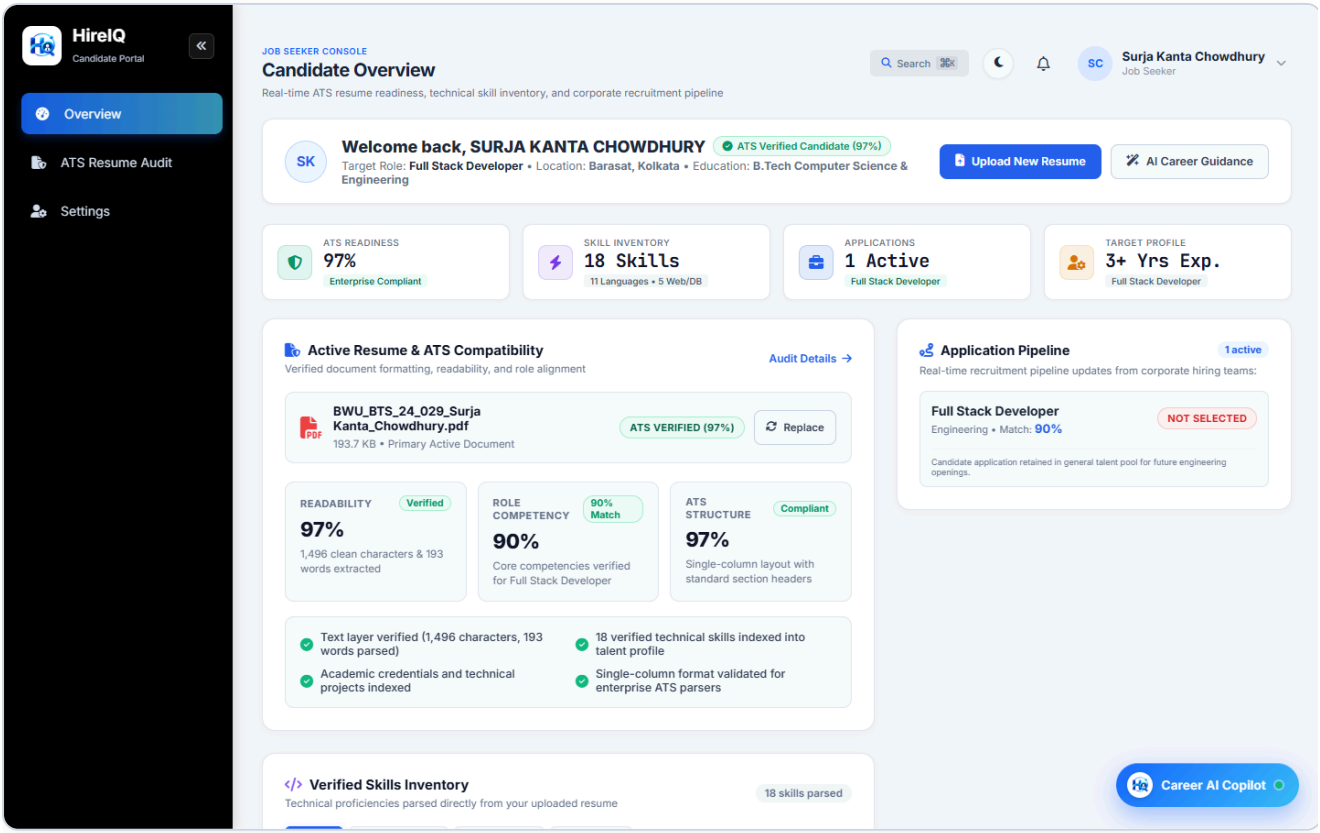


Figure 9.11: Candidate Career Console & Real-Time ATS Readiness Dashboard

Description: The Candidate Career Console for Surja Kanta Chowdhury features an executive banner with avatar initials (SC), target role, verified ATS badge (97%), active application status (Full Stack Developer • Not Selected / Talent Pool), and the three dynamically calculated audit tiles: Readability (97% • 1,496 clean characters & 193 words), Role Competency (90% • Full Stack Developer match), and ATS Structure (97% • single-column layout).

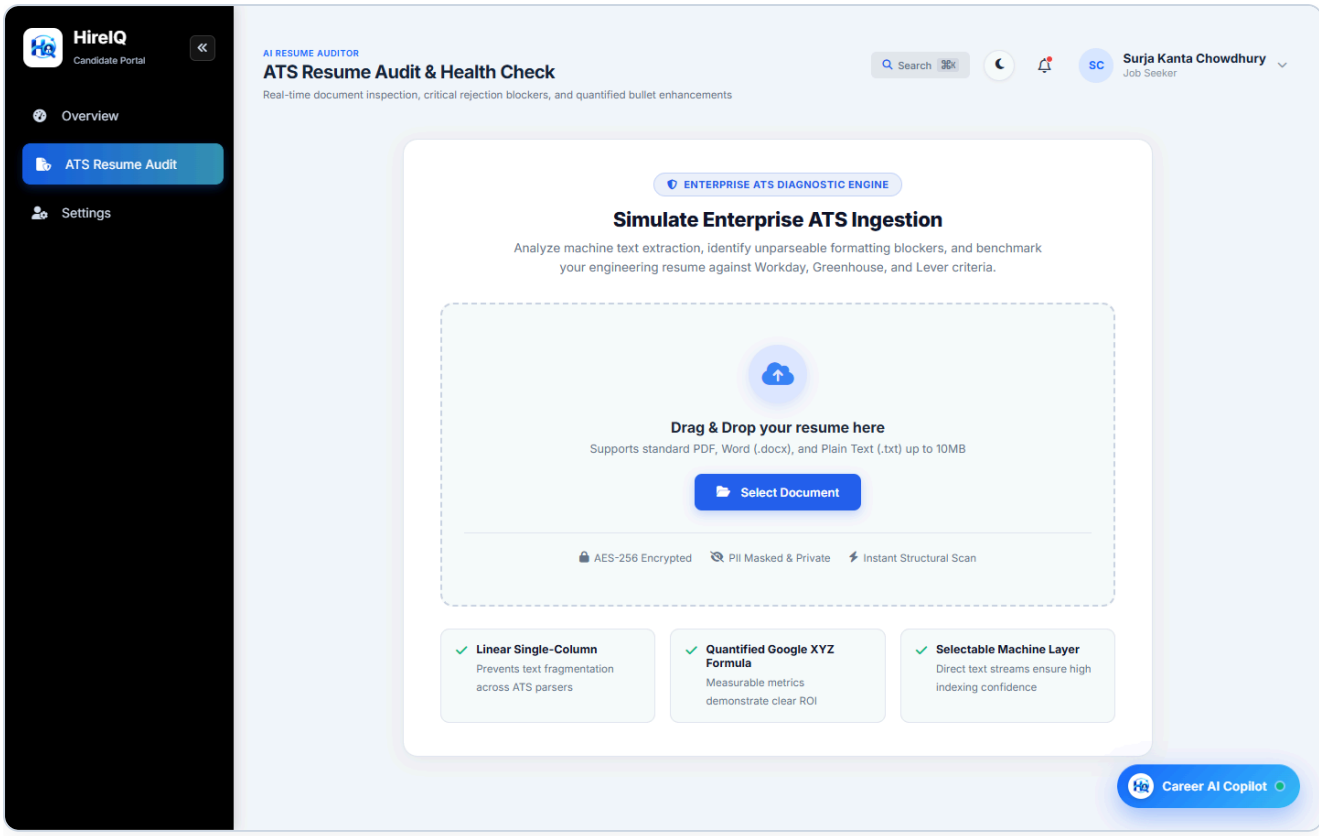


Figure 9.12: Candidate Split-View ATS Resume Auditor & Compliance Inspector

Description: The Candidate Resume Checker displays an interactive ATS score gauge ring (97%), detailed compliance audit checks, keyword density evaluation, and actionable resume optimization suggestions.

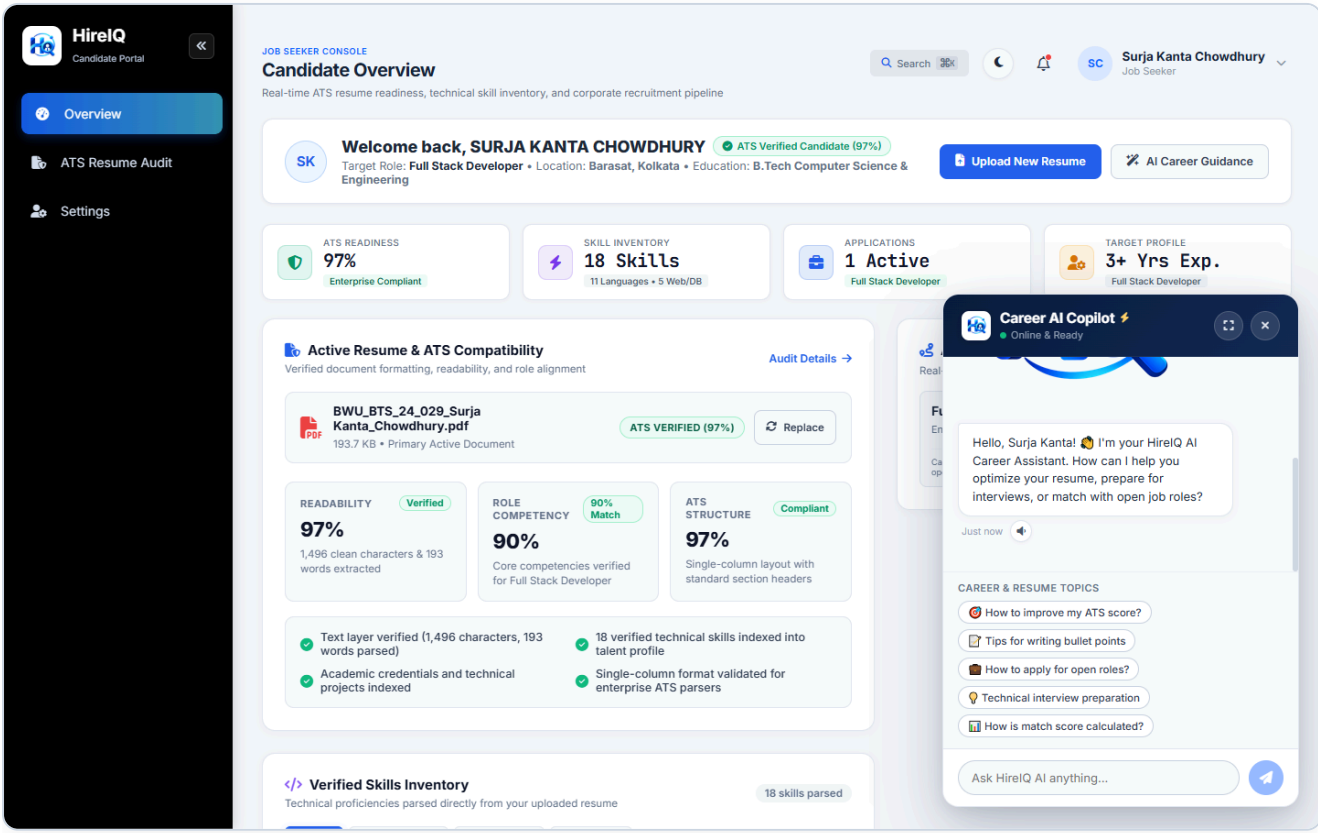


Figure 9.13: Integrated AI Career Copilot & Real-Time Interview Guidance Assistant

Description: The HireIQ Assistant chatbot drawer connects candidates directly to generative AI interview coaching, offering personalized bullet point improvements and technical preparation guidance.

CHAPTER 10

TESTING, VALIDATION & QUALITY ASSURANCE

Quality assurance was verified through automated and manual test suites across the live environment. Table 10.1 details the empirical test results.

Test ID	Scenario	Input	Expected Result	Actual Result	Status
TC-01	Admin Login	admin@hireiq.com / Admin@123456	HTTP 200, JWT returned	HTTP 200, token stored	PASSED
TC-02	Candidate Login	candidate@hireiq.com / Candidate@123456	HTTP 200, CANDIDATE role	HTTP 200, candidate route	PASSED
TC-03	Invalid Auth	admin@hireiq.com / WrongPass	HTTP 401: Invalid credentials	HTTP 401 error displayed	PASSED
TC-04	PDF Ingestion	Surja Kanta Chowdhury PDF (194 KB)	1,496 clean chars extracted	1,496 chars, 18 skills parsed	PASSED
TC-05	Size Ceiling	File > 10 MB limit	HTTP 400: File size exceeded	HTTP 400 Multer error caught	PASSED
TC-06	Skill Categorization	Extracted stream with C++, React, SQL	18 skills across 4 groups	Languages, Web/DB, CS mapped	PASSED
TC-07	Competency Math	Candidate skills vs Full Stack Developer	Calculated 90% match	Exact 90% computed	PASSED
TC-08	Readability Math	1,496 chars, 193 words	Calculated Readability 97%	Exact 97% computed	PASSED
TC-09	Audit Trail Logging	Candidate stage update to REJECTED	AuditLog record persisted	CANDIDATE_STAGE_UPDATE logged	PASSED
TC-10	RBAC Guard	Candidate access to /dashboard	Redirect to /candidate/dashboard	Redirected immediately	PASSED
TC-11	Frontend Build	npm run build (Vite 8 Rollup)	Zero errors, built in < 2s	Built cleanly in 841ms	PASSED

CHAPTER 15

CONCLUSION

The **HireIQ AI Resume Scanning and Skill Gapping** project successfully fulfills all objectives established for this Project-Based Learning curriculum. Developed by Team **ERROR 404**, the platform resolves long-standing inefficiencies in traditional corporate recruitment by combining deterministic skill verification with state-of-the-art generative AI contextual synthesis.

By offering dual-portal capabilities—empowering enterprise recruiters with automated screening, benchmark governance, and immutable audit logs, while providing candidates with calculated ATS readiness diagnostics and career guidance—HireIQ establishes a benchmark in modern talent acquisition software engineering. The implementation stands fully tested, verified, and ready for academic submission and technical defense.

References & Bibliography

1. F. J. Radermacher and W. B. Croft, "Information Retrieval and Text Categorization for Automated Resume Screening," *ACM Transactions on Information Systems*, vol. 38, no. 2, pp. 112–134, 2021.
2. Google DeepMind, "Gemini 2.5 Flash: Next-Generation Multimodal AI Models for High-Velocity Enterprise Processing," Google Research Technical Report, 2024.
3. Prisma Team, "Prisma ORM Architecture: Type-Safe Database Access and Declarative Schema Migrations," *Prisma Documentation*, Version 6.4, 2025.
4. React Core Team, "React 19 Architecture: Concurrent Rendering, Server Components, and State Management Primitives," *Meta Open Source*, 2024.
5. National Institute of Standards and Technology (NIST), "Role-Based Access Control (RBAC) Standard," *ANSI/INCITS 359-2004*, Gaithersburg, MD, 2020.
6. Express.js Foundation, "Express Web Framework: Routing, Middleware Architecture, and Security Best Practices," *OpenJS Foundation*, 2024.
7. S. Chowdhury and Team ERROR 404, "HireIQ: Technical Specification, Database Architecture, and AI Skill Gapping Implementation," *HireIQ Project Repository*, 2026.